Example

Week 2.2

- ogua han
  cifts
- Beh amar
- Yus Yonal

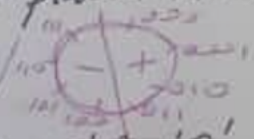
$72532 - 3250 = ?$

$$\begin{array}{r} 72532 \\ +\ 96750 \\ \hline \boxed{1}69282 \end{array} \Rightarrow +69282$$

carry

The left most bit shows the number is ng or positive if it is defined as SIGNED Number.

~~* To represent negative integers,~~ a bit is placed in the ~~leftmost position of the number.~~



The convention: 0 → shows pos., 1 → shows neg

~~* The convention is to make the sign bit '0' for positive and '1' for negative.~~

Addition of signed integers

* If the sign bits are the same add two magnitudes and give the sum the common sign.

* If the signs are different substract the smaller magnitude and give the larger magnitude's sign. (

Ex.s:

$$\begin{array}{rr} +6 & 00000110 \\ \pm\ +13 & 00001101 \\ \hline +19 & 00010011 \end{array}$$

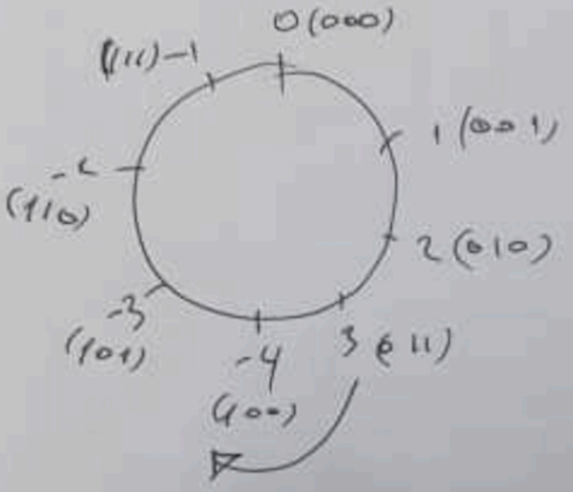
$$\begin{array}{rr} -6 & 11111010 \\ +\ +13 & +\ 00001101 \\ \hline +7 & \boxed{1}00000111 \\ \hline \end{array}$$

$$\begin{array}{r} 6 \\ -13 \\ \hline \end{array}$$

$$\begin{array}{r} -6 \\ -13 \\ \hline \end{array}$$

Over flows

- Add two positive numbers to get a negative number
- Add two negative numbers to get a positive number.



$$\begin{array}{rl} & 010 \rightarrow +2 \\ + & 011 \rightarrow +3 \quad \times \\ \hline & 101 \rightarrow (-3) \end{array}$$

overflow

$$\begin{array}{rl} & 110 \rightarrow -2 \\ + & 101 \rightarrow -3 \quad \times \\ \hline & \boxed{1}\boxed{011} \rightarrow +3 \end{array}$$

overflow

BINARY CODES

Decimal Symbol	BCD Digit
0	0000
1	0001
2	0010
3	0011
4	0100
5	0101
6	0110
7	0111
8	1000
9	1001

To represent a decimal number we will use 4

for each decimal digit!

for Ex., $396 \rightarrow \underbrace{0011}_{3}\ \underbrace{1001}_{9}\ \underbrace{0110}_{6}$

Each group of 4 bit represent a decimal digit.

* The binary combinations from 1010 to 1111 are not used and have no meaning in BCD.

Ex: Consider decimal 185 and its corresponding value in BCD and binary:

$(185)_{10} = (0001\ 1000\ 0101)_{BCD} = (10111001)_2$

BCD addition

$$\begin{array}{r} 4\\ +5\\ \hline 9\end{array} \qquad \begin{array}{r} 0100\\ +0101\\ \hline 1001\end{array} \qquad \begin{array}{r} 4\\ +8\\ \hline 12\end{array} \qquad \begin{array}{rl} 0100 & \\ +1000 & \\ \hline 1100 & \text{(it is greater than 9)}\\ +0110 & \text{(add 6)}\\ \hline \underbrace{1}_{1}\,\underbrace{0010}_{2} & \end{array}$$

$$\begin{array}{r} 8 \\ +\ 9 \\ \hline 17 \end{array} \qquad \begin{array}{r l} 1000 & \\ +\ 1001 & \\ \hline 10001 & \rightarrow \text{greater than } 9 \\ +\ 0110 & \rightarrow \text{add } 6 \\ \hline \underbrace{1}_{1}\,\underbrace{0111}_{7} & \end{array}$$

Example 184 + 576 = 760, in BCD:

$$\begin{array}{r r r l}
 & 1 \leftarrow & 1 \leftarrow & \\
 0001 & 1000 & 0100 & \rightarrow 184 \\
 +\ 0101 & 0111 & 0110 & \rightarrow 576 \\ \hline
 0111 & 10000 \rightarrow >9 & 1010 \rightarrow >9 & \\
 & 0110 & 0110 & \\ \hline
 & \textcircled{1}0110 & \textcircled{1}0000 & \\ \hline
 \underbrace{0111}_{7} & \underbrace{0110}_{6} & \underbrace{0000}_{0} &
\end{array}$$

Three ~~Four~~ Different Binary Codes for the decimal digits

Dec. Dig	BCD(8421)	2421		Excess-3	[illegible]
0		0000		0011	
1		0001		0100	
2		0010		0101	
3		0011		0110	
4		0100		0111	
5		1011	they are unused in BCD	1000	
6		1100		1001	
7		1101		1010	
8		1110		1011	
9		1111		1100	